

BAYANDELGER, Mongolia

WMO: 443520

Lat: 45.7303N

Lon: 112.3553E

Elev: 3612

StdP: 12.88

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-15.6	-24.3	1.6	-18.3	-21.0	2.0	-14.8	29.5	0.9	26.8	2.2	7.6	320	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	90.7	67.6	86.8	66.0	83.5	64.1	71.1	85.8	68.5	82.5	66.3	79.8	10.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	111.7	79.0	63.8	101.4	76.3	61.6	93.8	73.5	37.5	86.2	35.2	83.0	33.3	79.7	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.8	26.2	21.9	DB	-24.2	96.0	4.4	3.9	-27.4	98.8	-30.0	101.1	-32.4	103.2	-35.7	106.0	(4)
(5)			WB	-24.3	73.4	4.2	3.8	-27.3	76.1	-29.7	78.4	-32.1	80.5	-35.1	83.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	-1.5	7.0	23.8	41.4	54.8	65.9	71.3	68.0	56.4	39.4	17.9	3.0		
	DBStd	27.00	8.38	10.15	11.29	9.63	9.25	7.42	6.02	6.46	8.52	9.65	11.58	8.29		
	HDD50	6763	1597	1203	812	288	57	1	0	0	37	348	964	1456		
	HDD65	10540	2062	1623	1276	709	333	77	10	42	278	795	1414	1921		
	CDD50	2178	0	0	1	29	207	476	661	559	227	18	0	0		
	CDD65	480	0	0	0	0	18	103	205	135	19	0	0	0		
	CDH74	4717	0	0	0	5	282	1055	1966	1231	175	3	0	0		
	CDH80	1586	0	0	0	0	69	349	742	392	33	1	0	0		
	WSAvg	9.3	8.5	8.7	10.4	11.8	12.0	8.7	7.7	7.7	9.2	9.6	9.1	8.5		
	PrecAvg	7.2	0.0	0.0	0.1	0.2	0.7	1.6	2.0	1.3	0.6	0.3	0.2	0.1		
(15) Precipitation	PrecMax	14.8	0.1	0.2	0.4	1.3	2.8	4.2	6.3	4.3	2.4	1.1	0.4	0.2		
	PrecMin	3.6	0.0	0.0	0.0	0.0	0.0	0.3	0.4	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.7	0.0	0.0	0.1	0.3	0.8	0.9	1.5	1.0	0.6	0.3	0.1	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.3	37.2	58.6	73.5	86.6	93.5	96.7	94.0	83.9	69.6	48.6	28.1		
		MCWB	19.6	30.6	43.0	53.7	59.2	68.2	69.8	72.2	61.6	53.6	38.8	23.5		
	2%	DB	17.9	31.1	51.1	67.4	81.1	88.2	92.0	88.2	79.1	63.2	42.8	21.9		
		MCWB	15.2	26.6	38.6	49.5	57.4	64.9	68.7	68.2	59.6	49.6	34.9	19.0		
	5%	DB	13.8	26.3	45.2	62.6	76.5	84.2	87.6	84.1	74.7	58.1	38.0	17.9		
		MCWB	11.9	22.5	35.4	46.8	55.6	63.3	67.2	65.4	58.0	46.3	31.6	15.7		
	10%	DB	10.2	21.7	40.4	58.0	72.0	80.4	83.9	80.8	70.6	53.7	33.4	14.6		
		MCWB	8.8	18.8	32.1	44.1	53.2	61.6	65.6	63.4	55.5	43.2	28.4	12.8		
	0.4%	WB	20.0	31.4	43.7	56.5	62.5	70.7	76.5	74.9	64.5	56.2	40.2	23.9		
		MCDB	23.1	35.0	57.3	68.8	77.4	87.3	91.8	88.3	79.2	67.9	47.7	27.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	15.4	26.7	39.1	51.0	59.5	67.8	71.8	70.8	61.6	50.4	35.5	19.1		
		MCDB	17.8	31.1	50.2	65.4	77.7	83.5	86.5	85.0	76.3	61.6	43.0	21.6		
	5%	WB	12.0	22.8	35.8	47.5	56.9	65.5	69.4	67.8	59.1	46.9	32.3	15.8		
		MCDB	13.8	26.2	45.0	61.0	74.6	80.2	83.1	81.0	72.4	57.4	38.6	17.8		
	10%	WB	8.8	18.9	32.5	44.3	54.3	63.1	67.1	65.0	56.8	43.7	29.1	12.8		
		MCDB	10.2	21.7	40.3	57.7	70.1	76.7	80.3	77.7	68.7	53.2	34.0	14.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	16.8	19.7	21.7	22.8	19.9	19.3	19.8	20.3	18.8	16.0	13.7		
		MCDBR	16.3	20.5	24.1	26.9	28.5	23.8	23.8	23.8	24.4	23.2	19.9	16.0		
		MCWBR	14.9	17.0	16.6	16.9	15.5	12.3	11.0	12.2	13.8	15.4	14.7	14.0		
		MCDBR	16.4	20.4	23.6	25.8	26.4	21.7	21.5	22.3	22.5	22.6	19.3	15.7		
	5% WB	MCWBR	15.0	17.0	16.6	17.3	15.9	12.6	11.7	13.3	14.0	15.8	14.6	13.9		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.279	0.316	0.371	0.429	0.448	0.411	0.386	0.348	0.309	0.271	0.256		
	taud		2.336	2.284	2.233	2.121	1.989	2.044	2.221	2.259	2.301	2.345	2.349	2.307		
	Ebn at Noon		273	286	289	280	266	261	270	273	274	271	264	258		
	Edh at Noon		23	30	37	46	55	52	43	40	35	29	23	22		
(44) All-Sky Solar Radiation	RadAvg		664	1020	1500	1898	2123	2102	2039	1875	1538	1098	715	533		
	RadStd		54	62	59	81	109	127	80	80	70	62	54	42		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	-2.48	N/A	-1.22	-1.49	N/A	N/A	N/A	N/A

Nomenclature: See separate page